RAJALAKSHMI ENGINEERING COLLEGE

Vaibhav Kiran R 251801363-2025-AIDS V2

Started on	Tuesday, 25 August 2026, 12:36 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 1:14 PM
Time taken	37 mins 57 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void function(int n)
{
    int count = 0;

    int i = 1;
    count++; // Execution of: i = 1

    int s = 1;
    count++; // Execution of: s = 1

    count++; // First evaluation of while condition (s <= n)
    while (s <= n)
    {
        i++;
        count++; // Execution of: i++

        s += i;
        count++; // Execution of: s += i

        count++; // Re-evaluation of while condition (s <= n)
    }

    printf("%d\n", count);
}

int main()
{
    int n;
    if (scanf("%d", &n) == 1) {
```

```
32          function(n);
33      }
34      return 0;
35  }
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vaibhav Kiran R 251801363-2025-AIDS V2

Started on	Tuesday, 25 August 2026, 11:48 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:59 AM
Time taken	10 mins 33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, count = 0;

    scanf("%d", &n);

    if (n == 1)
    {
        count = 1;
    }
    else
    {
        for (int i = 1; i <= n; i++)
        {
            count += 5;
        }

        count += 2;
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vaibhav Kiran R 251801363-2025-AIDS V2

Started on	Tuesday, 25 August 2026, 12:05 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 12:09 PM
Time taken	4 mins 29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int num, i;
    int count = 0;

    scanf("%d", &num);

    count++;  // for loop initialization

    for (i = 1; i <= num; ++i)
    {
        count++;  // modulo operation
        count++;  // comparison

        if (num % i == 0)
        {
            count++;  // factor found / printf operation
        }
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔

	Input	Expected	Got	
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vaibhav Kiran R 251801363-2025-AIDS V2

Started on	Tuesday, 25 August 2026, 12:00 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 12:35 PM
Time taken	34 mins 48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    long long c = 0;

    scanf("%d", &n);

    c++;  // i = n/2

    for (int i = n / 2; i < n; i++)
    {
        c++;  // i < n

        for (int j = 1; j < n; j = 2 * j)
        {
            c++;  // j < n

            for (int k = 1; k < n; k = k * 2)
            {
                c++;  // k < n
                c++;  // c++
            }

            c++;  // final k < n
        }

        c++;  // final j < n
    }

    c++;  // final i < n

    printf("%lld", c);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Vaibhav Kiran R 251801363-2025-AIDS V2

Started on	Tuesday, 25 August 2026, 11:40 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:50 AM
Time taken	9 mins 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int count = 0;

    scanf("%d", &n);

    while (n != 0)
    {
        count += 4;
        n = n / 10;
    }

    count += 3;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course